

|        |   | Group                                                                                                                                  |                                                                                   |                                                                                 |                                                                                     |                                                                                |                                                                                  |                                                                                 |                                                                                |                                                                                  |                                                                                    |                                                                                   |                                                                                                             |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|--------|---|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
|        |   | 1                                                                                                                                      | 2                                                                                 |                                                                                 |                                                                                     |                                                                                |                                                                                  |                                                                                 |                                                                                |                                                                                  |                                                                                    |                                                                                   |                                                                                                             |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             | 3                                                                             | 4                                                                           | 5                                                                             | 6                                                                         | 7                                                                            | 0                                                                         |                                                                          |
| Period | 1 | <div> <div>Key</div> <div>Relative atomic mass</div> <div>Atomic symbol</div> <div>Name</div> <div>Atomic (proton number)</div> </div> |                                                                                   |                                                                                 |                                                                                     |                                                                                |                                                                                  |                                                                                 |                                                                                |                                                                                  |                                                                                    |                                                                                   |                                                                                                             |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             |                                                                               |                                                                             |                                                                               |                                                                           | <div> <div>1.0</div> <div>H</div> <div>Hydrogen</div> <div>1</div> </div>    |                                                                           | <div> <div>4.0</div> <div>He</div> <div>Helium</div> <div>2</div> </div> |
|        | 2 | <div> <div>6.9</div> <div>Li</div> <div>Lithium</div> <div>3</div> </div>                                                              | <div> <div>9.0</div> <div>Be</div> <div>Beryllium</div> <div>4</div> </div>       |                                                                                 |                                                                                     |                                                                                |                                                                                  |                                                                                 |                                                                                |                                                                                  |                                                                                    |                                                                                   |                                                                                                             |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             | <div> <div>10.8</div> <div>B</div> <div>Boron</div> <div>5</div> </div>       | <div> <div>12.0</div> <div>C</div> <div>Carbon</div> <div>6</div> </div>    | <div> <div>14.0</div> <div>N</div> <div>Nitrogen</div> <div>7</div> </div>    | <div> <div>16.0</div> <div>O</div> <div>Oxygen</div> <div>8</div> </div>  | <div> <div>19.0</div> <div>F</div> <div>Fluorine</div> <div>9</div> </div>   | <div> <div>20.2</div> <div>Ne</div> <div>Neon</div> <div>10</div> </div>  |                                                                          |
|        | 3 | <div> <div>23.0</div> <div>Na</div> <div>Sodium</div> <div>11</div> </div>                                                             | <div> <div>24.3</div> <div>Mg</div> <div>Magnesium</div> <div>12</div> </div>     |                                                                                 |                                                                                     |                                                                                |                                                                                  |                                                                                 |                                                                                |                                                                                  |                                                                                    |                                                                                   |                                                                                                             |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             | <div> <div>27.0</div> <div>Al</div> <div>Aluminium</div> <div>13</div> </div> | <div> <div>28.1</div> <div>Si</div> <div>Silicon</div> <div>14</div> </div> | <div> <div>31.0</div> <div>P</div> <div>Phosphorus</div> <div>15</div> </div> | <div> <div>32.1</div> <div>S</div> <div>Sulfur</div> <div>16</div> </div> | <div> <div>35.5</div> <div>Cl</div> <div>Chlorine</div> <div>17</div> </div> | <div> <div>39.9</div> <div>Ar</div> <div>Argon</div> <div>18</div> </div> |                                                                          |
|        | 4 | <div> <div>39.1</div> <div>K</div> <div>Potassium</div> <div>19</div> </div>                                                           | <div> <div>40.1</div> <div>Ca</div> <div>Calcium</div> <div>20</div> </div>       | <div> <div>45.0</div> <div>Sc</div> <div>Scandium</div> <div>21</div> </div>    | <div> <div>47.9</div> <div>Ti</div> <div>Titanium</div> <div>22</div> </div>        | <div> <div>50.9</div> <div>V</div> <div>Vanadium</div> <div>23</div> </div>    | <div> <div>52.0</div> <div>Cr</div> <div>Chromium</div> <div>24</div> </div>     | <div> <div>54.9</div> <div>Mn</div> <div>Manganese</div> <div>25</div> </div>   | <div> <div>55.8</div> <div>Fe</div> <div>Iron</div> <div>26</div> </div>       | <div> <div>58.9</div> <div>Co</div> <div>Cobalt</div> <div>27</div> </div>       | <div> <div>58.7</div> <div>Ni</div> <div>Nickel</div> <div>28</div> </div>         | <div> <div>63.5</div> <div>Cu</div> <div>Copper</div> <div>29</div> </div>        | <div> <div>63.5</div> <div>Zn</div> <div>Zinc</div> <div>30</div> </div>                                    | <div> <div>69.7</div> <div>Ga</div> <div>Gallium</div> <div>31</div> </div>    | <div> <div>72.6</div> <div>Ge</div> <div>Germanium</div> <div>32</div> </div>    | <div> <div>74.9</div> <div>As</div> <div>Arsenic</div> <div>33</div> </div>   | <div> <div>79.0</div> <div>Se</div> <div>Selenium</div> <div>34</div> </div>   | <div> <div>79.9</div> <div>Br</div> <div>Bromine</div> <div>35</div> </div>   | <div> <div>83.8</div> <div>Kr</div> <div>Krypton</div> <div>36</div> </div> |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|        | 5 | <div> <div>85.5</div> <div>Rb</div> <div>Rubidium</div> <div>37</div> </div>                                                           | <div> <div>87.6</div> <div>Sr</div> <div>Strontium</div> <div>38</div> </div>     | <div> <div>88.9</div> <div>Y</div> <div>Yttrium</div> <div>39</div> </div>      | <div> <div>91.2</div> <div>Zr</div> <div>Zirconium</div> <div>40</div> </div>       | <div> <div>92.9</div> <div>Nb</div> <div>Niobium</div> <div>41</div> </div>    | <div> <div>95.9</div> <div>Mo</div> <div>Molybdenum</div> <div>42</div> </div>   | <div> <div>(98)</div> <div>Tc</div> <div>Technetium</div> <div>43</div> </div>  | <div> <div>101.1</div> <div>Ru</div> <div>Ruthenium</div> <div>44</div> </div> | <div> <div>102.9</div> <div>Rh</div> <div>Rhodium</div> <div>45</div> </div>     | <div> <div>106.4</div> <div>Pd</div> <div>Palladium</div> <div>46</div> </div>     | <div> <div>107.9</div> <div>Ag</div> <div>Silver</div> <div>47</div> </div>       | <div> <div>112.4</div> <div>Cd</div> <div>Cadmium</div> <div>48</div> </div>                                | <div> <div>114.8</div> <div>In</div> <div>Indium</div> <div>49</div> </div>    | <div> <div>118.7</div> <div>Sn</div> <div>Tin</div> <div>50</div> </div>         | <div> <div>121.8</div> <div>Sb</div> <div>Antimony</div> <div>51</div> </div> | <div> <div>127.6</div> <div>Te</div> <div>Tellurium</div> <div>52</div> </div> | <div> <div>126.9</div> <div>I</div> <div>Iodine</div> <div>53</div> </div>    | <div> <div>131.3</div> <div>Xe</div> <div>Xenon</div> <div>54</div> </div>  |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|        | 6 | <div> <div>132.9</div> <div>Cs</div> <div>Caesium</div> <div>55</div> </div>                                                           | <div> <div>137.3</div> <div>Ba</div> <div>Barium</div> <div>56</div> </div>       | <div> <div>138.9</div> <div>La*</div> <div>Lanthanum</div> <div>57</div> </div> | <div> <div>178.5</div> <div>Hf</div> <div>Hafnium</div> <div>72</div> </div>        | <div> <div>180.9</div> <div>Ta</div> <div>Tantalum</div> <div>73</div> </div>  | <div> <div>183.8</div> <div>W</div> <div>Tungsten</div> <div>74</div> </div>     | <div> <div>186.2</div> <div>Re</div> <div>Rhenium</div> <div>75</div> </div>    | <div> <div>190.2</div> <div>Os</div> <div>Osmium</div> <div>76</div> </div>    | <div> <div>192.2</div> <div>Ir</div> <div>Iridium</div> <div>77</div> </div>     | <div> <div>195.1</div> <div>Pt</div> <div>Platinum</div> <div>78</div> </div>      | <div> <div>197.0</div> <div>Au</div> <div>Gold</div> <div>79</div> </div>         | <div> <div>200.6</div> <div>Hg</div> <div>Mercury</div> <div>80</div> </div>                                | <div> <div>204.4</div> <div>Tl</div> <div>Thallium</div> <div>81</div> </div>  | <div> <div>207.2</div> <div>Pb</div> <div>Lead</div> <div>82</div> </div>        | <div> <div>209.0</div> <div>Bi</div> <div>Bismuth</div> <div>83</div> </div>  | <div> <div>(209)</div> <div>Po</div> <div>Polonium</div> <div>84</div> </div>  | <div> <div>(210)</div> <div>At</div> <div>Astatine</div> <div>85</div> </div> | <div> <div>(222)</div> <div>Rn</div> <div>Radon</div> <div>86</div> </div>  |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|        | 7 | <div> <div>(223)</div> <div>Fr</div> <div>Francium</div> <div>87</div> </div>                                                          | <div> <div>(226)</div> <div>Ra</div> <div>Radium</div> <div>88</div> </div>       | <div> <div>(227)</div> <div>Ac*</div> <div>Actinium</div> <div>89</div> </div>  | <div> <div>(261)</div> <div>Rf</div> <div>Rutherfordium</div> <div>104</div> </div> | <div> <div>(262)</div> <div>Db</div> <div>Dubnium</div> <div>105</div> </div>  | <div> <div>(266)</div> <div>Sg</div> <div>Seaborgium</div> <div>106</div> </div> | <div> <div>(264)</div> <div>Bh</div> <div>Bohrium</div> <div>107</div> </div>   | <div> <div>(277)</div> <div>Hs</div> <div>Hassium</div> <div>108</div> </div>  | <div> <div>(268)</div> <div>Mt</div> <div>Mitnerium</div> <div>109</div> </div>  | <div> <div>(271)</div> <div>Ds</div> <div>Darmstadtium</div> <div>110</div> </div> | <div> <div>(272)</div> <div>Rg</div> <div>Roentgenium</div> <div>111</div> </div> | <div> <div>Elements with atomic numbers 112–116 have been reported but not fully authenticated</div> </div> |                                                                                |                                                                                  |                                                                               |                                                                                |                                                                               |                                                                             |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|        |   | <div> <div>140.1</div> <div>Ce</div> <div>Cerium</div> <div>58</div> </div>                                                            | <div> <div>140.9</div> <div>Pr</div> <div>Praseodymium</div> <div>59</div> </div> | <div> <div>144.2</div> <div>Nd</div> <div>Neodymium</div> <div>60</div> </div>  | <div> <div>144.9</div> <div>Pm</div> <div>Promethium</div> <div>61</div> </div>     | <div> <div>150.4</div> <div>Sm</div> <div>Samarium</div> <div>62</div> </div>  | <div> <div>152.0</div> <div>Eu</div> <div>Europium</div> <div>63</div> </div>    | <div> <div>157.2</div> <div>Gd</div> <div>Gadolinium</div> <div>64</div> </div> | <div> <div>158.9</div> <div>Tb</div> <div>Terbium</div> <div>65</div> </div>   | <div> <div>162.5</div> <div>Dy</div> <div>Dysprosium</div> <div>66</div> </div>  | <div> <div>164.9</div> <div>Ho</div> <div>Holmium</div> <div>67</div> </div>       | <div> <div>167.3</div> <div>Er</div> <div>Erbium</div> <div>68</div> </div>       | <div> <div>168.9</div> <div>Tm</div> <div>Thulium</div> <div>69</div> </div>                                | <div> <div>173.0</div> <div>Yb</div> <div>Ytterbium</div> <div>70</div> </div> | <div> <div>175.0</div> <div>Lu</div> <div>Lutetium</div> <div>71</div> </div>    |                                                                               |                                                                                |                                                                               |                                                                             |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |
|        |   | <div> <div>232.0</div> <div>Th</div> <div>Thorium</div> <div>90</div> </div>                                                           | <div> <div>(231)</div> <div>Pa</div> <div>Protactinium</div> <div>91</div> </div> | <div> <div>238.1</div> <div>U</div> <div>Uranium</div> <div>92</div> </div>     | <div> <div>(237)</div> <div>Np</div> <div>Neptunium</div> <div>93</div> </div>      | <div> <div>(242)</div> <div>Pu</div> <div>Plutonium</div> <div>94</div> </div> | <div> <div>(243)</div> <div>Am</div> <div>Americium</div> <div>95</div> </div>   | <div> <div>(247)</div> <div>Cm</div> <div>Curium</div> <div>96</div> </div>     | <div> <div>(245)</div> <div>Bk</div> <div>Berkelium</div> <div>97</div> </div> | <div> <div>(251)</div> <div>Cf</div> <div>Californium</div> <div>98</div> </div> | <div> <div>(254)</div> <div>Es</div> <div>Einsteinium</div> <div>99</div> </div>   | <div> <div>(253)</div> <div>Fm</div> <div>Fermium</div> <div>100</div> </div>     | <div> <div>(256)</div> <div>Md</div> <div>Mendelevium</div> <div>101</div> </div>                           | <div> <div>(254)</div> <div>No</div> <div>Nobelium</div> <div>102</div> </div> | <div> <div>(257)</div> <div>Lr</div> <div>Lawrencium</div> <div>103</div> </div> |                                                                               |                                                                                |                                                                               |                                                                             |                                                                               |                                                                             |                                                                               |                                                                           |                                                                              |                                                                           |                                                                          |

# PERIODICITY

## ATOMIC RADIUS AND IONISATION ENERGY

Name: .....

## THE STRUCTURE OF THE PERIODIC TABLE (GROUPS & PERIODS)

The Periodic Table has been developed over time by many chemists including Johann Döbereiner, John Newlands, Dmitri Mendeleev, Henry Moseley and Glenn Seaborg. The purpose of the Periodic Table is to arrange the many elements in three main ways:

### Atomic (Proton) Number

Elements are arranged in the Periodic Table by **increasing atomic (proton) number** e.g. nitrogen is placed directly after carbon as an atom of carbon has six protons in the nucleus and an atom of nitrogen has seven protons in the nucleus.

### Periods

Periods are the **horizontal rows of elements** that show trends (gradual changes) in physical and chemical properties across the period. These trends are \_\_\_\_\_ across each period e.g. across Period 2 the elements change from metal to non-metal (*via* the metalloid boron) and this pattern is repeated across Period 3 where the elements also change from metal to non-metal (*via* the metalloid silicon).

*Periodicity*



### Groups

Groups are the **vertical columns of elements** that have similar physical and chemical properties. Atoms in each group have the same number of outer shell electrons resulting in similar properties e.g. atoms of the elements in Group 7 have seven outer shell electrons and react by gaining one electron to form 1- anions.

### s-, p-, d- and f- Blocks

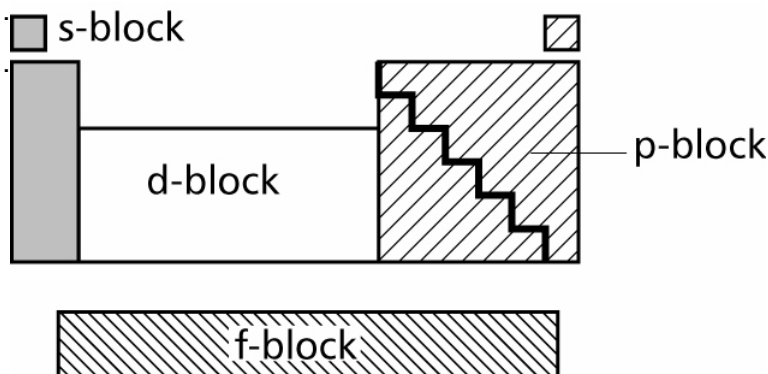
The Periodic Table is also structured in blocks of elements, linked to sub-shells e.g. magnesium, Mg would be described as an s-block element because the **highest energy sub-shell is an s-subshell** ( $1s^2 2s^2 p^6 3s^2$ ).



**Exercise 1** Write the electron configuration for the following atoms and hence determine, and explain, what block it would be found in.

a) Si .....

b) Sc .....

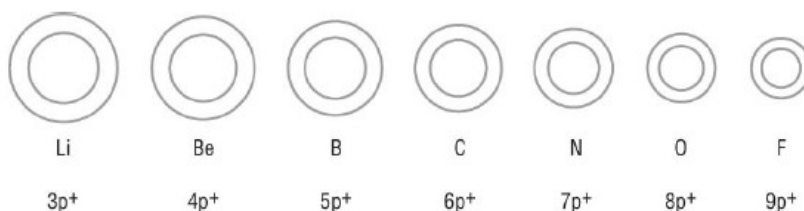


## PERIODICITY OF PHYSICAL PROPERTIES OF ELEMENTS

Remember... **Periodicity is the repeating pattern of trends across different periods** (as you move down the groups).



At AS level you have to be able to describe and explain trends in atomic radius, first ionisation energies, and melting & boiling points in terms of periodicity (across the period and down the groups).



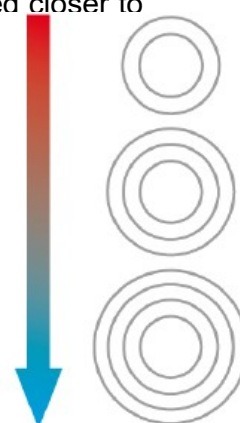
### Atomic Radius

#### Across a Period

- Atomic radii (size of atom) \_\_\_\_\_.
- Outermost electrons in the \_\_\_\_\_ shell.
- Nuclear charge increases (more \_\_\_\_\_).
- Nuclear attraction on outermost electrons \_\_\_\_\_ (electrons are pulled closer to the nucleus).

#### Down a Group

- Atomic radii (size of atom) \_\_\_\_\_.
- More \_\_\_\_\_.
- Greater distance of outermost electron to nucleus.
- More \_\_\_\_\_ on outermost electron from inner electrons.
- Nuclear attraction on outermost electrons \_\_\_\_\_ (electrons are not pulled as close to nucleus).
- Although nuclear charge \_\_\_\_\_ this is 'far outweighed' by **more** distance and shielding.



### First Ionisation Energies

Across a period first ionisation energy shows a general \_\_\_\_\_.

Down a group first ionisation energy \_\_\_\_\_.

You have already studied the trends in ionisation energies across a period and down a group. Exercise 2 will revise the key points of ionisation energies.



## Exercise 2

1. Define the term *first ionisation energy*.

.....  
.....[1]

2. Write an equation, with state symbols, to represent the **first ionisation energy** of carbon.

.....[1]

3. List the **three** factors which influence ionisation energy.

.....  
.....  
.....[3]

4. **State** and **explain** the **general trend** in first ionisation energy:

a) **across** a period.

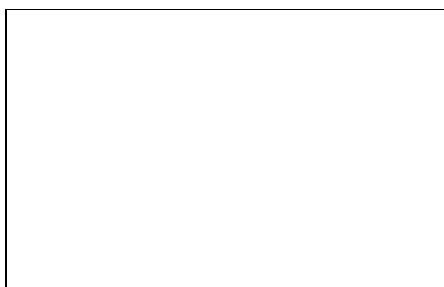
.....  
.....  
.....[3]

b) **down** a group.

.....  
.....  
.....  
.....[4]

## Electronegativity

**Task:** Draw a 'dot-and-cross-' diagram to show the covalent bonding in  $\text{Cl}_2$ .



When the covalently bonded atoms are the same type or have relatively the same electronegativities, a permanent dipole cannot be created. However, as the bonding shared pair of electrons are moving all of the time, there will be a \_\_\_\_\_.

When covalently bonded atoms have the same or almost the same electronegativities and a temporary dipole arises, it results in a \_\_\_\_\_-\_\_\_\_\_ bond.

|                                                                                   |    | Electronegativity increases                                                        |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |  |
|-----------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|--|
| Period                                                                            |    |  |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |  |
|  | 1  | H<br>2.20                                                                          |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            |            | He         |  |
|                                                                                   | 2  | Li<br>0.98                                                                         | Be<br>1.57 |            |            |            |            |            |            |            |            |            |            | B<br>2.04  | C<br>2.55  | N<br>3.04  | O<br>3.44  | F<br>3.98  | Ne         |  |
|                                                                                   | 3  | Na<br>0.93                                                                         | Mg<br>1.31 |            |            |            |            |            |            |            |            |            |            | Al<br>1.61 | Si<br>1.90 | P<br>2.19  | S<br>2.58  | Cl<br>3.16 | Ar         |  |
|                                                                                   | 4  | K<br>0.82                                                                          | Ca<br>1.00 | Sc<br>1.36 | Ti<br>1.54 | V<br>1.63  | Cr<br>1.66 | Mn<br>1.55 | Fe<br>1.83 | Co<br>1.88 | Ni<br>1.91 | Cu<br>1.90 | Zn<br>1.65 | Ga<br>1.81 | Ge<br>2.01 | As<br>2.18 | Se<br>2.55 | Br<br>2.96 | Kr<br>3.00 |  |
|                                                                                   | 5  | Rb<br>0.82                                                                         | Sr<br>0.95 | Y<br>1.22  | Zr<br>1.33 | Nb<br>1.6  | Mo<br>2.16 | Tc<br>1.9  | Ru<br>2.2  | Rh<br>2.28 | Pd<br>2.20 | Ag<br>1.93 | Cd<br>1.69 | In<br>1.78 | Sn<br>1.96 | Sb<br>2.05 | Te<br>2.1  | I<br>2.66  | Xe<br>2.6  |  |
|                                                                                   | 6  | Cs<br>0.79                                                                         | Ba<br>0.89 | *          | Hf<br>1.3  | Ta<br>1.5  | W<br>2.36  | Re<br>1.9  | Os<br>2.2  | Ir<br>2.20 | Pt<br>2.28 | Au<br>2.54 | Hg<br>2.00 | Tl<br>1.62 | Pb<br>2.33 | Bi<br>2.02 | Po<br>2.0  | At<br>2.2  | Rn         |  |
|                                                                                   | 7  | Fr<br>0.7                                                                          | Ra<br>0.9  | **         | Rf         | Db         | Sg         | Bh         | Hs         | Mt         | Ds         | Rg         | Uub        | Uut        | Uuq        | Uup        | Uuh        | Uus        | Uuo        |  |
| Lanthanides                                                                       | *  | La<br>1.1                                                                          | Ce<br>1.12 | Pr<br>1.13 | Nd<br>1.14 | Pm<br>1.13 | Sm<br>1.17 | Eu<br>1.2  | Gd<br>1.2  | Tb<br>1.1  | Dy<br>1.22 | Ho<br>1.23 | Er<br>1.24 | Tm<br>1.25 | Yb<br>1.1  | Lu<br>1.27 |            |            |            |  |
| Actinides                                                                         | ** | Ac<br>1.1                                                                          | Th<br>1.3  | Pa<br>1.5  | U<br>1.38  | Np<br>1.36 | Pu<br>1.28 | Am<br>1.13 | Cm<br>1.28 | Bk<br>1.3  | Cf<br>1.3  | Es<br>1.3  | Fm<br>1.3  | Md<br>1.3  | No<br>1.3  | Lr<br>1.3  |            |            |            |  |

The three most electronegative elements of the Periodic Table are: \_\_\_\_, \_\_\_\_ and \_\_\_\_.

\_\_\_\_ is the most electronegative.

### Exercise 1

Predict whether each of the following bonds will be polar or non-polar and write them in the appropriate box. Write a dipole across the polar bonds.

H-F; F-F; O=O; H-O; C=O; C-Cl; C=C; N-H; B-F; Br-Br; C-Br; H-Br; P-F; C-H.

| Non-Polar Bond | Polar Covalent Bond |
|----------------|---------------------|
|                |                     |

**Q.** Which of the bonds above will have the greatest permanent dipole? Why?



## Questions



1. The atomic radii of the elements Li to F and Na to Cl are shown in the table below.

|                           |       |       |       |       |       |       |       |
|---------------------------|-------|-------|-------|-------|-------|-------|-------|
| <b>ELEMENT</b>            | Li    | Be    | B     | C     | N     | O     | F     |
| <b>ATOMIC RADIUS / nm</b> | 0.134 | 0.125 | 0.090 | 0.077 | 0.075 | 0.073 | 0.071 |
| <b>ELEMENT</b>            | Na    | Mg    | Al    | Si    | P     | S     | Cl    |
| <b>ATOMIC RADIUS / nm</b> | 0.154 | 0.145 | 0.130 | 0.118 | 0.110 | 0.102 | 0.099 |

a) Using only the elements in this table, select:

i) An element with both metallic and non-metallic properties.

.....[1]

ii) The element with the **largest** first ionisation energy.

.....[1]

iii) An element with a giant molecular/covalent structure.

.....[1]

b) Explain what causes the general **decrease** in atomic radii across each period.

.....  
 .....  
 .....  
 .....[3]

c) Predict and explain whether a sodium **ion** is larger, smaller or the same size as a sodium atom.

.....  
 .....  
 .....  
 .....[3]

d) Explain why a nitrogen atom is **larger** than an oxygen atom.

.....  
 .....  
 .....  
 .....[3]

2. In this question you are provided with information about ionisation energies of elements. You are also provided with some additional information that will help you answer part b).

a) Define the term *first ionisation energy*.

.....  
 .....  
 .....  
 .....[1]

b) The table shown below provides data on elements in Period 2 of the Periodic Table.

| ELEMENT                                                  | Li          | Be          | B                | C                | N                |
|----------------------------------------------------------|-------------|-------------|------------------|------------------|------------------|
| NUMBER OF PROTONS                                        | 3           | 4           | 5                | 6                | 7                |
| ELECTRON CONFIGURATION                                   | $1s^2 2s^1$ | $1s^2 2s^2$ | $1s^2 2s^2 2p^1$ | $1s^2 2s^2 2p^2$ | $1s^2 2s^2 2p^3$ |
| 1 <sup>st</sup> IONISATION ENERGY / kJ mol <sup>-1</sup> | 520         | 900         | 801              | 1086             | 1402             |

Describe and explain the **general** trend in first ionisation energies shown by the elements Li-N.

.....  
 .....  
 .....  
 .....[4]

3. In the Periodic Table, **describe** and **explain** the trend in **atomic radii** shown by the Group 2 elements Be-Ra.

.....  
 .....  
 .....  
 .....[4]

4. The Periodic Table is a chemist's way of ordering the elements to show patterns of chemical and physical properties. In the Periodic Table, elements are arranged in order of their atomic number. State and explain the trend in chemical reactivity down Group 2. In your answer, use ideas about ionisation energy.

.....  
 .....  
 .....

.....[4]

5. The compounds  $\text{NH}_3$ ,  $\text{BF}_3$  and  $\text{HI}$  all have covalent bonding and simple molecular structures. The Pauling electronegativity values shown below can be used to predict the polarity in these compounds.

|     |
|-----|
| H   |
| 2.1 |

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| Li  | Be  | B   | C   | N   | O   | F   |
| 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 3.5 | 4.0 |
| Na  |     |     |     |     |     | Cl  |
| 0.9 |     |     |     |     |     | 3.0 |
| K   |     |     |     |     |     | Br  |
| 0.8 |     |     |     |     |     | 2.8 |
|     |     |     |     |     |     | I   |
|     |     |     |     |     |     | 2.5 |

a) The electronegativity values can be used to predict the polarity of a bond. Copy and complete the boxes below, show the polarity of each bond by adding  $\delta^+$  or  $\delta^-$  to each bond. The first box has been completed for you.

|                                       |     |     |     |
|---------------------------------------|-----|-----|-----|
| $\delta^- \text{O}-\text{H} \delta^+$ | H-N | F-B | H-I |
|---------------------------------------|-----|-----|-----|

[3]